

Year 7 · Chapter 10 · Analysing data: answers (teacher copy)

10.01 Interpreting graphs (booklet p. 3)

Examples (model these live)

E1 How many students chose Netball?
scale goes up in 2s · 8 students

E2 How many more chose Cricket than Soccer?
Cricket 10, Soccer 6 · $10 - 6 = 4$

E3 What fraction of the students chose Soccer?
total $6 + 8 + 4 + 10 = 28$ · $\frac{6}{28} = \frac{3}{14}$

Questions (17 on today's skill, 7 from earlier lessons)

- | | | | |
|-------------------------------------|--|--|---|
| 1 2 | 9 $\frac{3}{20} = \frac{15}{100} = 15\%$ 4.12 | 15 10 out of 28 · $10 \div 28 \times 100 \approx 36\%$ | 20 Walk is 25% (a quarter) · $120 \div 4 = 30$ students |
| 2 Netball | 10 $6 + 14 + 4 + 2 = 26$ | 16 Walk is 25% (a quarter) · $80 \div 4 = 20$ students | 21 5.5 symbols · $5.5 \times 10 = 55$ books |
| 3 25% | 11 warmest: Jun (28°C) · $28 - 19 = 9^\circ\text{C}$ | 17 1 L = 1000 mL · $\frac{250}{1000} = \frac{1}{4}$ 4.12 | 22 test $x = 1, 2$ and 3 in each · $y = 4x + 3$ 9.05 |
| 4 Cricket | 12 25% 4.12 | 18 4 symbols · $4 \times 5 = 20$ books | 23 6 out of 22 · $6 \div 22 \times 100 \approx 27\%$ |
| 5 12.5% | 13 Cricket 14, Soccer 8 · $14 - 8 = 6$ | 19 (0, -5) 9.06 | 24 4% 4.12 |
| 6 8 | 14 $\frac{23}{25} \times 100\% = 92\%$ 4.12 | | |
| 7 Cricket 4, Soccer 2 · $4 - 2 = 2$ | | | |
| 8 $8 + 6 + 10 + 8 = 32$ | | | |

10.02 Misleading graphs (booklet p. 5)

Examples (model these live)

E1 Why is this graph misleading?
the scale starts at 50, not 0 · small differences look big

E2 Bar B looks 3 times as tall as bar A. Is B really 3 times A?
A is 52, B is 56 · no: only 4 more

E3 A scale goes 0, 10, 20, 50, 100 with equal gaps. What is wrong?
the gaps are not equal steps · uneven scale

Questions (18 on today's skill, 6 from earlier lessons)

- | | | | |
|--|--|--|--|
| 1 at 100; the difference looks bigger than it is | 7 looks: $12 \div 6 = 2$ times · really: $92 \div 86 \approx 1.07$ times | 13 Netball 10.01 | 19 5 9.04 |
| 2 The area grows much faster than the number, so it looks far bigger. | 8 The steps are not equal, so the bars are not in proportion. | 14 looks: $12 \div 4 = 3$ times · really: $92 \div 84 \approx 1.1$ times | 20 No. The rise is only 1 thousand (about 4%); the cut scale makes it look huge. |
| 3 at 200; the difference looks bigger than it is | 9 Cricket 14, Netball 8 · $14 - 8 = 6$ 10.01 | 15 Start the vertical scale at 0 (with even steps), so the bars are in proportion. | 21 Start the vertical scale at 0 (with even steps), so the bars are in proportion. |
| 4 at 100; the difference looks bigger than it is | 10 looks: $12 \div 4 = 3$ times · really: $52 \div 44 \approx 1.18$ times | 16 No. The rise is only 3 thousand (about 11%); the cut scale makes it look huge. | 22 D → C, 3 → 2 · C2 9.01 |
| 5 looks: $16 \div 4 = 4$ times · really: $96 \div 84 \approx 1.14$ times | 11 looks: $18 \div 4 = 4.5$ times · really: $98 \div 84 \approx 1.17$ times | 17 $6 + 8 + 10 + 4 = 28$ 10.01 | 23 No. The rise is only 3 thousand (about 14%); the cut scale makes it look huge. |
| 6 at 200; the difference looks bigger than it is | 12 Start the vertical scale at 0 (with even steps), so the bars are in proportion. | 18 Start the vertical scale at 0 (with even steps), so the bars are in proportion. | 24 Cricket 10.01 |

10.03 Dot plots (booklet p. 7)

Examples (model these live)

E1 How many values are in the dot plot?
count every dot · 12

E2 Which value is the most common?
tallest stack · 4

E3 Draw a dot plot of 1, 3, 3, 4, 4, 4, 6.
number line from 1 to 6 · one dot for each value

Questions (18 on today's skill, 8 from earlier lessons)

- | | | | |
|---------------------------|---|---|--|
| 1 12 | 9 dots: 4×1, 5×1, 6×1, 7×2, 8×1, 9×2 | 15 dots: 2×1, 3×1, 4×1, 5×3, 8×1, 9×1 | 21 $70 - 6 = 64 \text{ cm}^2$ 8.08 |
| 2 8 | 10 yes: 10 | 16 Cricket 12, Swim 8 · $12 - 8 = 4$ 10.01 | 22 8 out of 10 · 80% |
| 3 10 | 11 warmest: May (30°C) · $29 - 19 = 10^\circ\text{C}$ 10.01 | 17 most values between 2 and 8 · cluster around 5 | 23 most values between 2 and 7 · cluster around 6 |
| 4 13 | 12 yes: 10 | 18 10 out of 10 · 100% | 24 $x: 1 + 1 = 2$, $y: 4 + 6 = 10$ · (2, 10) 9.03 |
| 5 4 | 13 10 out of 11 · $\frac{10}{11}$ | 19 $2 + 8 + 4 + 12 = 26$ 10.01 | 25 12 out of 14 · $\frac{12}{14}$ |
| 6 1 | 14 at 100; the difference looks bigger than it is 10.02 | 20 most values between 2 and 8 · cluster around 5.5 | 26 The area grows much faster than the number, so it looks far bigger. 10.02 |
| 7 12 | | | |
| 8 200 cm^2 8.07 | | | |

10.04 Stem-and-leaf plots (booklet p. 9)

Examples (model these live)

E1 Write the values in the second row.
stem 3, leaves 2, 5, 7 · 32, 35, 37

E2 How many values are in the plot?
count the leaves · 7

E3 Draw an ordered stem-and-leaf plot of 43, 27, 35, 41, 29, 38.

stems 2, 3, 4 · 2 | 7 9, 3 | 5 8, 4 | 1 3

Questions (17 on today's skill, 6 from earlier lessons)

- | | | | |
|---|--|---|---|
| 1 42, 43 | 8 4 | 14 looks: $14 \div 2 = 7$ times · really: $74 \div 62 \approx 1.19$ times 10.02 | 18 10 out of 10 passed · 100% |
| 2 8 | 9 $2 + 10 + 14 + 12 = 38$ 10.01 | 15 8 out of 10 passed · 80% | 19 3, 6, 9 9.05 |
| 3 26 and 55 | 10 3 | 16 stem 2 · most values are in that ten | 20 stem 3 · most values are in that ten |
| 4 9 | 11 $2 4 7 \cdot 3 1 2 7 8 \cdot 4 8 \cdot 5 4$ | 17 dots: $2 \times 1, 3 \times 3, 4 \times 1, 8 \times 2, 9 \times 1$ 10.03 | 21 8 out of 10 passed · 80% |
| 5 13 and 52 | 12 4 10.03 | | 22 $2 \times 1 - 3 = -1$ · yes 9.06 |
| 6 32, 38, 39 | 13 7 out of 10 passed · 70% | | 23 7 |
| 7 $2 2 7 8 \cdot 3 2 4 9 \cdot 4 \cdot 5 0 6$ | | | |

10.05 Mean and mode (booklet p. 11)

Examples (model these live)

- | | | |
|---|---|---|
| E1 Find the mean of 4, 7, 9, 6, 9.
total 35 , 5 values · $35 \div 5 = 7$ | E2 Find the mode of 4, 7, 9, 6, 9.
9 occurs twice · mode 9 | E3 The mean of 5 numbers is 8. Four of them are 6, 9, 7, 10. Find the fifth.
total $5 \times 8 = 40$ · $40 - 32 = 8$ |
|---|---|---|

Questions (35 on today's skill, 12 from earlier lessons)

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|---------------------------------------|--|---|--|
| 1 9 | 16 104; 26 | 30 $154 \div 6 \approx 25.7$ | 39 new total: $5 \times 83 = 415$ · $415 - 324 = 91$ |
| 2 4 | 17 79 3.03 | 31 You cannot tell what is being measured or in what units. 10.02 | 40 Soccer 10, Swim $8 \cdot 10 - 8 = 2$ 10.01 |
| 3 92; 23 | 18 8 | 32 mean: $35 \div 5 = 7$ · mode: 3 and 10 | 41 new total: $5 \times 83 = 415$ · $415 - 324 = 91$ |
| 4 8 | 19 total: $156 \cdot 156 \div 6 = 26$ | 33 total: $5 \times 11 = 55$ · $55 - 43 = 12$ | 42 mean: 8 · new mean: 18 (10 more) |
| 5 6 | 20 $161 \div 6 \approx 26.8$ | 34 new total: $5 \times 75 = 375$ · $375 - 292 = 83$ | 43 25 7.10 |
| 6 100; 25 | 21 $6 + 10 + 10 + 8 = 34$ 10.01 | 35 $4 \times 2 + 4 = 12$ · yes 9.06 | 44 total: $5 \times 8 = 40$ · $40 - 36 = 4$ |
| 7 6 | 22 mean: $30 \div 5 = 6$ · mode: 8 | 36 mean: 8 · new mean: 18 (10 more) | 45 Swim 10.01 |
| 8 4 | 23 total: $224 \cdot 224 \div 7 = 32$ | 37 4 3.03 | 46 mean: $35 \div 5 = 7$ · mode: 8 |
| 9 4 10.03 | 24 $172 \div 6 \approx 28.7$ | 38 total: $5 \times 15 = 75$ · $75 - 62 = 13$ | 47 new total: $5 \times 78 = 390$ · $390 - 304 = 86$ |
| 10 2 | 25 mean: $30 \div 5 = 6$ · mode: 2 | | |
| 11 120; 30 | 26 28 ha 8.04 | | |
| 12 6 | 27 total: $308 \cdot 308 \div 7 = 44$ | | |
| 13 26 m 8.03 | 28 12.5% 10.01 | | |
| 14 total: $216 \cdot 216 \div 6 = 36$ | 29 total: $5 \times 10 = 50$ · $50 - 45 = 5$ | | |
| 15 5 | | | |

10.06 Median and range (booklet p. 13)

Examples (model these live)

- | | | |
|--|--|---|
| E1 Find the median of 8, 3, 9, 5, 6.
in order: 3, 5, 6, 8, 9 · median 6 | E2 Find the median of 4, 10, 7, 2.
in order: 2, 4, 7, 10 · $(4 + 7) \div 2 = 5.5$ | E3 Find the range of 12, 25, 18, 9, 21.
largest 25, smallest 9 · $25 - 9 = 16$ |
|--|--|---|

Questions (34 on today's skill, 12 from earlier lessons)

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|--|---|--|--|
| 1 10 | 18 in order: 3, 4, 9, 20, 23, 27, 27, 29 · median: 21.5 | 28 $24 - \text{smallest} = 10$ · smallest = 14 | 39 mean 27.6, median 22 · median: the outlier 93 pulls the mean up |
| 2 11 | 19 in order: 11, 17, 34, 42, 52, 53, 54 · median 42, range 43 | 29 in order: 12, 17, 19, 32, 34, 44, 55 · median 32, range 43 | 40 mean 26.1, median 18 · median: the outlier 78 pulls the mean up |
| 3 1, 12, 12, 14, 22, 24, 24 | 20 $6 + 2 + 12 + 14 = 34$ 10.01 | 30 76; 19 10.05 | 41 $(8 + \square) \div 2 = 9.5$ · $\square = 11$ |
| 4 13 | 21 highest 3, lowest $-6 \cdot 3 - (-6) = 9^\circ\text{C}$ | 31 highest 2, lowest $-3 \cdot 2 - (-3) = 5^\circ\text{C}$ | 42 6 cm^2 8.04 |
| 5 36 | 22 in order: 11, 12, 16, 17, 25, 26, 28 · median: 21 | 32 $2 1 \cdot 3 6 8 9 \cdot 4 0 3 3 7$ 10.04 | 43 $27 - \text{smallest} = 17$ · smallest = 10 |
| 6 1, 2, 8, 24, 28, 28 | 23 in order: 14, 29, 34, 36, 48 · median 34, range 34 | 33 $29 - \text{smallest} = 19$ · smallest = 10 | 44 at 50; the difference looks bigger than it is 10.02 |
| 7 6 | 24 highest 4, lowest $-7 \cdot 4 - (-7) = 11^\circ\text{C}$ | 34 mean 21.1, median 13 · median: the outlier 78 pulls the mean up | 45 highest 3, lowest $-8 \cdot 3 - (-8) = 11^\circ\text{C}$ |
| 8 33 | 25 102 mL 8.10 | 35 $42 = b \times 7$ · $b = 6 \text{ m}$ 8.07 | 46 mean 22.9, median 11 · median: the outlier 87 pulls the mean up |
| 9 7 10.05 | 26 in order: 10, 13, 16, 26, 27, 29 · median: 21 | 36 $(8 + \square) \div 2 = 11$ · $\square = 14$ | |
| 10 26 | 27 You cannot tell what is being measured or in what units. 10.02 | 37 6 10.03 | |
| 11 10, 14, 19, 21, 23, 25 | | 38 $19 - \text{smallest} = 7$ · smallest = 12 | |
| 12 14 | | | |
| 13 17 7.10 | | | |
| 14 in order: 3, 5, 11, 23, 29, 29 · median: 17 | | | |
| 15 32 | | | |
| 16 5, 10, 10, 13, 22, 23, 23 | | | |
| 17 10 10.04 | | | |

10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 15)

Examples (model these live)

E1 Find the range and the mode.
range $8 - 2 = 6$ · mode 4

E2 Find the median.
9 values: the 5th · median 4

E3 Find the mean.
total 39 · $39 \div 9 \approx 4.3$

Questions (19 on today's skill, 8 from earlier lessons)

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|--------------------------------|----------------------------------|---|---|
| 1 8 | 10 7 values: the 4th · median 44 | 16 The area grows much faster than the number, so it looks far bigger. 10.02 | 21 mean $320 \div 8 = 40$ · median 40.5, range 35 |
| 2 6 | 11 $6 + 12 + 2 + 2 = 22$ 10.01 | 17 mean $336 \div 8 = 42$ · median 41.5, range 27 | 22 92; 23 10.05 |
| 3 $52 - 31 = 21$ | 12 9 values: the 5th · median 38 | 18 mean $360 \div 8 = 45$ · median 47, range 30 | 23 range $6 \rightarrow 8$; median 5.5 $\rightarrow 6$ · mode: 8 $\rightarrow 8$ |
| 4 7 | 13 total 40, 10 values · mean 4 | 19 range $5 \rightarrow 8$; median 5.5 $\rightarrow 6$ · mode: 7 $\rightarrow 7$ | 24 mean $304 \div 8 = 38$ · median 40.5, range 20 |
| 5 4 | 14 12 cm^2 8.12 | 20 \$3.12 7.10 | 25 0.462 L 8.13 |
| 6 $50 - 23 = 27$ | 15 9 values: the 5th · median 3 | | 26 total 50, 10 values · mean 5 |
| 7 5 | | | 27 50% 10.01 |
| 8 C4 9.01 | | | |
| 9 9 values: the 5th · median 5 | | | |

10.08 Comparing data (booklet p. 17)

Examples (model these live)

E1 Class A: 5, 7, 8, 8, 9. Class B: 3, 6, 8, 9, 10. Compare the medians.
A: 8, B: 8 · the same

E2 Compare the ranges. Which class is more consistent?
A: 4, B: 7 · A (smaller range)

E3 Which class did better overall? Use the mean.
A: $37 \div 5 = 7.4$, B: $36 \div 5 = 7.2$ · A, just

Questions (31 on today's skill, 9 from earlier lessons)

- | | | | |
|---|---|--|---|
| 1 A (16 and 9) | 15 A: 56, B: 61 · B has the higher mean | 23 A: 66, B: 51 · A has the higher mean | 31 $F = 5$, $V = 6$, $E = 9 \cdot 5 + 6 - 9 = 2$ ✓ 8.09 |
| 2 A (9 and 7) | 16 25 10.06 | 24 5, 6, 7 9.05 | 32 medians: A 41, B 38 · ranges: A 17, B 24 |
| 3 A (18 and 12) | 17 A: 56, B: 57 · B has the higher mean | 25 medians: A 40, B 45 · ranges: A 23, B 21 | 33 68; 17 10.05 |
| 4 A (14 and 7) | 18 ranges: A 2, B 11 · A: a smaller range means less spread | 26 The area grows much faster than the number, so it looks far bigger. 10.02 | 34 A: mean 4, range 5 · B: mean 4, range 5 |
| 5 B (12 and 16) | 19 $10 + 14 + 12 + 12 = 48$ 10.01 | 27 ranges: A 2, B 11 · A: a smaller range means less spread | 35 A: mean 4, range 5 · B: mean 4, range 5 |
| 6 B (12 and 13) | 20 A: 65, B: 62 · A has the higher mean | 28 A: 55, B: 61 · B has the higher mean | 36 medians: A 44, B 41 · ranges: A 29, B 20 |
| 7 A (13 and 12) | 21 ranges: A 2, B 11 · A: a smaller range means less spread | 29 medians: A 39, B 34 · ranges: A 26, B 32 | 37 2.621 L 8.13 |
| 8 A (12 and 8) | 22 ranges: A 2, B 11 · A: a smaller range means less spread | 30 A: mean 3, range 5 · B: mean 3, range 4 | 38 A: mean 2, range 2 · B: mean 4, range 4 |
| 9 B (7 and 12) | | | 39 9 10.04 |
| 10 A (10 and 8) | | | 40 medians: A 26, B 39 · ranges: A 26, B 32 |
| 11 A: 59, B: 63 · B has the higher mean | | | |
| 12 68 7.10 | | | |
| 13 B (8 and 13) | | | |
| 14 B (13 and 19) | | | |